

ARKA, Russian Federation

WMO: 249880

Lat: 60.083N

Lon: 142.333E

Elev: 650

StdP: 14.35

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.1	-37.3	-46.2	0.4	-40.9	-42.4	0.5	-36.9	14.5	11.5	13.5	12.2	0.9	10	0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.3	(c) 79.3	(d) 61.7	(e) 75.3	(f) 59.8	(g) 71.3	(h) 58.3	(i) 63.9	(j) 75.0	(k) 61.6	(l) 71.8	(m) 59.7	(n) 68.2	(o) 4.5	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	78.7	67.3	57.7	72.7	63.5	56.1	68.7	61.9	29.6	75.1	27.8	72.6	26.5	68.1	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.2	10.4	9.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-45.7	87.6	4.1	4.0	-48.7	90.5	-51.1	92.9	-53.4	95.2	-56.4	98.1
			-45.6	68.2	3.8	3.3	-48.3	70.6	-50.5	72.5	-52.6	74.4	-55.4	76.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	-13.1	-7.5	5.8	23.9	40.2	52.6	58.3	57.2	44.7	24.3	1.1	-10.7
	DBStd	27.99	14.39	12.32	10.75	8.74	6.19	6.53	5.09	4.67	6.71	10.21	13.61	13.50
	HDD50	10411	1955	1609	1370	784	309	46	2	5	184	797	1467	1883
	HDD65	15258	2420	2029	1835	1234	768	373	218	244	610	1262	1917	2348
	CDD50	641	0	0	0	0	6	124	258	229	24	0	0	0
	CDD65	13	0	0	0	0	0	1	9	3	0	0	0	0
	CDH74	459	0	0	0	0	5	146	223	84	1	0	0	0
	CDH80	95	0	0	0	0	1	32	51	11	0	0	0	0
Wind	WSAvg	3.6	3.7	3.5	3.6	3.6	3.8	3.7	3.2	3.0	3.2	3.8	4.2	4.4
Precipitation	PrecAvg	17.6	0.4	0.3	0.4	1.0	1.6	2.0	3.0	2.9	2.7	1.7	0.9	0.6
	PrecMax	35.2	2.2	1.1	1.6	2.7	4.7	4.4	14.5	8.6	6.6	4.5	3.5	2.9
	PrecMin	10.0	0.0	0.0	0.0	0.0	0.1	0.2	0.6	0.5	0.7	0.1	0.0	0.0
	PrecStd	4.7	0.5	0.3	0.5	0.7	1.0	1.0	2.4	1.8	1.6	1.1	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.8	27.6	36.9	48.6	71.6	85.2	86.5	80.9	69.4	50.1	35.3	26.7
		MCWB	22.8	24.1	28.4	37.1	52.1	62.4	65.4	63.3	54.9	40.1	32.3	21.7
	2%	DB	21.6	23.5	31.7	44.5	64.2	78.3	80.1	75.9	64.2	44.6	31.4	21.7
		MCWB	18.9	20.0	26.6	34.5	48.3	59.2	63.7	60.8	51.1	37.0	29.3	18.7
	5%	DB	16.9	18.9	27.8	41.1	59.6	73.7	75.7	72.5	60.2	40.7	27.8	17.1
		MCWB	14.3	15.6	23.0	32.6	45.8	57.4	61.3	59.6	50.0	34.6	25.8	14.2
	10%	DB	11.3	13.7	24.6	37.9	55.0	68.6	71.3	68.5	56.9	37.6	23.1	11.8
		MCWB	9.2	10.9	20.3	30.7	43.5	55.2	59.2	58.2	49.6	32.5	21.0	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.9	30.4	37.7	52.9	63.7	69.2	66.5	57.2	42.3	33.4	22.7
		MCDB	26.6	27.4	33.6	47.0	70.0	79.6	84.1	74.6	64.0	46.9	35.1	25.9
	2%	WB	19.1	20.0	26.7	35.1	49.2	60.7	65.4	63.1	55.0	38.5	30.0	18.6
		MCDB	21.5	23.3	31.6	42.9	62.7	75.6	76.9	72.6	60.0	43.6	31.5	21.7
	5%	WB	14.3	15.5	23.4	33.1	46.2	58.2	62.5	61.0	53.1	35.6	25.7	14.1
		MCDB	16.9	18.7	27.7	39.6	57.6	71.3	73.0	69.0	57.4	39.1	27.7	16.9
	10%	WB	9.2	11.0	20.4	31.5	43.9	56.0	60.3	59.3	50.9	33.6	21.0	9.4
		MCDB	11.5	13.7	24.5	36.9	54.3	67.1	69.3	66.0	55.1	36.4	23.0	11.6
Mean Daily Temperature Range	5% DB	MDBR	18.8	25.8	28.9	21.4	20.5	21.7	18.3	18.5	19.7	17.6	16.7	15.7
		MCDBR	14.1	18.4	27.6	24.0	32.3	35.4	31.1	28.9	26.8	19.7	9.7	14.0
		MCWBR	12.6	16.0	22.3	15.0	17.2	17.1	15.2	14.8	15.0	12.6	8.5	12.5
	5% WB	MCDBR	14.4	17.7	25.0	21.1	30.0	31.0	27.6	23.7	19.3	16.5	8.3	13.5
		MCWBR	13.0	15.4	20.1	13.5	16.7	16.0	14.6	13.2	11.6	11.2	7.7	12.2
Clear-Sky Solar Irradiance	taub		0.219	0.265	0.325	0.440	0.459	0.426	0.471	0.426	0.340	0.300	0.239	0.196
	taud		2.059	2.099	1.983	1.724	1.882	2.215	2.105	2.223	2.434	2.234	2.048	1.962
	Ebn at Noon		179	231	246	232	241	255	238	240	249	215	161	135
	Edh at Noon		14	24	38	60	56	41	44	37	25	21	14	10
All-Sky Solar Radiation	RadAvg		153	484	1059	1608	1662	1689	1481	1207	852	505	211	79
	RadStd		11	25	48	105	164	159	218	144	115	52	21	5

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page